

# PIPE-Cypher: Automatic Enterprise Benchmark Generation for Text-to-Cypher Systems

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## Abstract

Enterprises increasingly use property graphs and Cypher to analyze fraud, risk, identity, customer, and operational data. Public Text2Cypher benchmarks rarely reflect private schemas, terminology, access patterns, or difficulty distributions. We present PIPE-Cypher, a local-model pipeline for generating organization-specific natural-language-to-Cypher benchmarks. PIPE-Cypher combines schema introspection, reverse-query grounding, constrained Cypher generation, deterministic read-only and schema validation, execution feedback, repair, diversity controls, and LLM-judge review. The system is designed for industry settings where benchmark generation must avoid paid APIs, preserve data governance, and remain repeatable as graphs evolve.

## 1 Introduction

Property graphs encode enterprise relationships directly: account transfers, identity entitlements, access paths, customer interactions, and fraud rings. Cypher is a common query language for these graphs. As LLMs become natural-language interfaces for graph analytics, organizations need reliable benchmarks for natural-language-to-Cypher systems on their own schemas.

Static public benchmarks are insufficient for this setting. They cannot contain private labels, relationship types, categorical values, or domain-specific wording, and they do not track schema evolution. PIPE-Cypher addresses this gap by generating private, repeatable, execution-grounded NL-Cypher benchmarks.

## 2 Related Work

Recent Text2Cypher resources, including Mind the Query [1], SyntheT2C [10], Auto-Cypher [8], and Text2Cypher [5], show that high-quality Cypher data requires more than asking an LLM for query pairs. These works motivate schema grounding, execution checks, synthetic generation, verification, and complexity-aware evaluation. PIPE-Cypher differs by targeting private enterprise benchmark generation as the primary artifact: organizations bring their own graph, run local models, enforce Cypher constraints, and receive an executable benchmark with diversity and judge metadata.

Text-to-SQL benchmarks such as Spider 2.0 [2] and BIRD [3] motivate realistic database tasks, execution-based evaluation, and enterprise workflow complexity. CIKM AutoQuery [9] motivates strategy-level workload generation analysis and separating workload quality from model quality. LDBC FinBench [7] and SNB [6] provide graph workloads suitable for industry-oriented evaluation.

For generated benchmark quality, PIPE-Cypher combines text-generation diversity diagnostics with text-to-query structure metrics. Distinct-n [4] and self-BLEU-style redundancy checks [11] capture lexical repetition, while schema coverage, query-signature diversity, structural feature rates, and normalized entropy over graph/category/difficulty cells capture properties that matter for executable Cypher benchmarks.

| Gate                              | Evidence checked                                           | Failure mode caught                       |
|-----------------------------------|------------------------------------------------------------|-------------------------------------------|
| Read-only safety                  | blocked Cypher write/admin tokens                          | destructive or operational queries        |
| Schema validity                   | labels, relationship types, properties, categorical values | hallucinated graph vocabulary or values   |
| Direction validity                | observed relationship directions                           | reversed or invalid traversals            |
| Cypher analysis and normalization | structure extraction, rewrite safety, RETURN DISTINCT      | unsafe rewrites or duplicate/noisy rows   |
| Question constraints              | quoted values use exact literals                           | fuzzy matching of exact user values       |
| Execution                         | query runs and returns rows                                | syntactic validity without answerability  |
| LLM judge                         | ambiguity, semantic alignment, schema use                  | executable but semantically weak examples |

Table 1: PIPE-Cypher candidate acceptance gates.

### 3 Method

PIPE-Cypher has five stages: schema profiling, template generation, reverse Cypher grounding, constrained Cypher generation and repair, deterministic validation and execution, and LLM-judge review. Deterministic checks enforce read-only safety, syntax shape, schema-visible labels and properties, explicit relationship types, observed relationship directions, schema-provided categorical values, and non-empty execution where required. A lightweight Cypher analyzer extracts return aliases, variables, labels, relationship observations, risky constructs, and rewrite skip reasons so normalization is auditable rather than a silent string edit. The direction gate interprets both outgoing and incoming Cypher arrow syntax before schema checking, and rejects undirected relationship patterns so directionality is an executable constraint rather than only a prompt instruction.

### 4 Implementation

The implementation is a Python package with Neo4j as the experimental backend. The paper frames the contribution around Cypher and property graphs rather than a single vendor. Local models are served through a vLLM/OpenAI-compatible endpoint on ds-serv6, using Qwen3.5 models for generation and judging.

The intended full-quality model is Qwen3.5-35B-A3B, but our June 1, 2026 live evidence uses the cached Qwen3.5-9B fallback. The 35B-A3B weights were staged on ds-serv6 under root-backed storage. A serving-capacity check estimated that the staged weights require four A5000 GPUs under our vLLM budget, while only one GPU was safely free in the live snapshot, so the reported generation and downstream results use the 9B fallback.

The built-in FinBench profile is grounded in the public datagen snapshot export and includes typed node and relationship properties. Live schema inspection also records low-cardinality string values as categorical constraints for prompting and validation. The generated Neo4j import script uses uniqueness constraints for nodes and creates, rather than merges, transaction relationships so repeated account-to-account events are preserved. The SNB reference profile is grounded in the official Neo4j/Cypher headers and read-query files.

For reproducible smoke runs, PIPE-Cypher can use seeded templates with deterministic reverse-binding queries. Full runs use a mixed mode that prepends these proven workload seeds to LLM-proposed templates, then records accepted and rejected candidates for yield analysis. Retrieved few-shot examples replace graph-specific values with typed placeholders, preserving query structure while reducing tenant-value leakage and memorized entity reuse. A dependency-light value grounder adds typed prompt annotations for categorical values and reverse-bound entities, covering punctuation variants, possessives, plurals, synonyms, name partials, and small typos.

| Graph    | Target/category | Binding limit | Min seed capacity | All categories meet target |
|----------|-----------------|---------------|-------------------|----------------------------|
| FinBench | 250             | 300           | 300               | Yes                        |
| SNB      | 125             | 200           | 200               | Yes                        |

Table 2: Built-in seed-template capacity after removing the reverse-binding 10-row cap and adding slotted negation/ranking seeds. Capacity is a theoretical scale-readiness check; final examples still must pass validation, execution, diversity, and judge gates.

| Setting                | Value                               |
|------------------------|-------------------------------------|
| Accepted examples      | 3,000                               |
| Primary graph          | LDBC FinBench, 2,000 examples       |
| Secondary graph        | LDBC SNB, 1,000 examples            |
| Categories             | 8 balanced categories, 375 each     |
| Generation/judge model | Local Qwen3.5-9B fallback           |
| Execution backend      | Neo4j Community, two live databases |
| Judge audit packet     | 80 sampled accepted/rejected pairs  |

Table 3: Full live experimental setup used for the generated benchmark artifact.

For LLM-judge review, the implementation slices schema context to the labels, relationship types, and properties used by the candidate query. This preserves validation against the full schema while keeping local 9B smoke-model prompts within context limits on larger schemas.

The deterministic Cypher layer borrows from production lessons in the BalkanID Cypher system: schema-only prompting, exact matching, relationship direction discipline, `RETURN DISTINCT`, reserved variable rejection, categorical values, required contextual return columns, fuzzy value annotations, placeholderized retrieval examples, and parser-aware rewrite boundaries. PIPE-Cypher records parser-style structure features and skips rewrites for risky constructs such as `UNION`, `CALL`, `UNWIND`, `WHERE EXISTS`, multiple `WHERE` clauses, or reserved variables.

We also estimate seed-template capacity before full generation. This check caught and fixed a scale blocker: reverse-binding execution was hard-capped to 10 rows, and no-slot negation/ranking seeds could not support full category targets. PIPE-Cypher now uses the configured binding limit and includes slotted negation and ranking seeds for both graphs.

## 5 Experiments

The generated benchmark contains 3,000 accepted examples: 2,000 from LDBC FinBench and 1,000 from LDBC SNB. Categories include simple retrieval, complex retrieval, simple aggregation, complex aggregation, boolean existence, negation/difference, path/temporal transaction, and ranking/top-k.

| Graph    | Candidates | Accepted | Acceptance | Categories at target |
|----------|------------|----------|------------|----------------------|
| FinBench | 3,404      | 2,000    | 0.588      | 8/8                  |
| SNB      | 1,373      | 1,000    | 0.728      | 8/8                  |
| Total    | 4,777      | 3,000    | 0.628      | 16/16                |

Table 4: Full live generation with local Qwen3.5-9B. Candidate counts include the initial sequential run and category-specific recovery top-ups.

| Metric                 | Offline | Live FinBench | Live SNB |
|------------------------|---------|---------------|----------|
| Records generated      | 4       | 4             | 4        |
| Accepted records       | 4       | 4             | 4        |
| Read-only pass         | 4       | 4             | 4        |
| Syntax-valid pass      | 4       | 4             | 4        |
| Schema-valid pass      | 4       | 4             | 4        |
| Execution-success pass | 4       | 4             | 4        |
| Judge pass             | 4       | 4             | 4        |

Table 5: Smoke evidence. The live runs used loaded Neo4j graphs and local Qwen3.5-9B judge review; these numbers verify end-to-end wiring but are not final experimental results.

## 6 Results

The full live run produced 3,000 accepted examples from 4,777 candidates using local Qwen3.5-9B for generation and judging. Category-specific recovery top-ups filled the only under-target categories from the initial sequential run. Every exported example passed read-only, syntax, schema, execution, non-empty result, and judge gates.

As an implementation smoke check, we generated and loaded FinBench SF0.1 into Neo4j, loaded the official SNB Cypher test-data into a second Neo4j instance, introspected both live schemas, served Qwen3.5-9B locally through vLLM, and ran four-category smokes on both graphs. All eight accepted examples passed read-only, syntax, schema, execution, non-empty result, and LLM-judge gates. A broader seeded FinBench smoke additionally accepted 8/8 examples across all planned categories with four easy and four medium examples.

We also ran a small live mini-ablation. A FinBench LLM-only probe accepted 0/16 candidates, with deterministic validation tagging all 16 as generic unlabeled node scans. Re-enabling seeded templates, scalar reverse-binding, duplicate-question control, deterministic fallback, execution validation, and LLM-judge review accepted 16/29 FinBench candidates, with two accepted examples in every planned category. The analogous SNB mixed run accepted 8/8 candidates.

We additionally materialized the planned baseline and ablation configurations and ran target-five suites on both FinBench and SNB. The strict unconstrained local-LLM baseline disables seeded template fallback, retrieval, rewrite, repair, deterministic Cypher fallback, and the LLM judge; it produced no usable records

| Live run                | Records | Accepted | Acceptance |
|-------------------------|---------|----------|------------|
| FinBench LLM-only probe | 16      | 0        | 0.000      |
| FinBench mixed mini     | 29      | 16       | 0.552      |
| SNB mixed mini          | 8       | 8        | 1.000      |

Table 6: Live mini-ablation evidence with local Qwen3.5-9B. The LLM-only probe disables deterministic seed/fallback, while mixed runs combine seeded templates with LLM-proposed templates and full validation/judge gates.

| Setting           | Graph    | Records | Accepted | Acceptance | Categories at target |
|-------------------|----------|---------|----------|------------|----------------------|
| Unconstrained LLM | FinBench | 0       | 0        | 0.000      | 0/8                  |
| Unconstrained LLM | SNB      | 0       | 0        | 0.000      | 0/8                  |
| Reverse-only      | FinBench | 40      | 40       | 1.000      | 8/8                  |
| Reverse-only      | SNB      | 40      | 40       | 1.000      | 8/8                  |
| Validators+repair | FinBench | 40      | 40       | 1.000      | 8/8                  |
| Validators+repair | SNB      | 40      | 40       | 1.000      | 8/8                  |
| No retrieval      | FinBench | 41      | 40       | 0.976      | 8/8                  |
| No retrieval      | SNB      | 40      | 40       | 1.000      | 8/8                  |
| No rewrite        | FinBench | 41      | 40       | 0.976      | 8/8                  |
| No rewrite        | SNB      | 40      | 40       | 1.000      | 8/8                  |
| No LLM judge      | FinBench | 40      | 40       | 1.000      | 8/8                  |
| No LLM judge      | SNB      | 40      | 40       | 1.000      | 8/8                  |
| Full PIPE-Cypher  | FinBench | 41      | 40       | 0.976      | 8/8                  |
| Full PIPE-Cypher  | SNB      | 40      | 40       | 1.000      | 8/8                  |

Table 7: Live target-five ablation evidence with local Qwen3.5-9B. Each graph run targets 5 accepted examples per category.

on either graph because the local model did not return parseable template JSON. Once reverse grounding and seeded workload structure are available, the small target saturates across variants, which is why the full results emphasize scale, recovery, and export quality rather than only small-yield differences.

We then ran a live mid-scale generation pass with a target of five accepted examples per category for each graph. FinBench accepted 40/46 candidates and SNB accepted 40/47 candidates; both runs reached the category target in all eight planned categories.

The accepted full-run records were exported into a benchmark package with stable example identifiers, train/dev/test splits, result samples, gate metadata, aggregate statistics, and a manifest hash for artifact tracking. The export contains 3,000 accepted examples: 2,000 FinBench, 1,000 SNB, and 375 accepted examples in every planned category.

We report diversity as a diagnostic, not only as a design target. The full export has perfect category balance and near-perfect difficulty balance, but the Qwen3.5-9B fallback run still has low query-signature diversity because seeded graph-grounded templates are doing substantial work. This is useful evidence for industry deployment: the artifact is balanced and executable, while the appendix makes template concentration visible rather than hiding it.

| Live run           | Records | Accepted | Acceptance | Categories at target |
|--------------------|---------|----------|------------|----------------------|
| FinBench mid-scale | 46      | 40       | 0.870      | 8/8                  |
| SNB mid-scale      | 47      | 40       | 0.851      | 8/8                  |

Table 8: Live mid-scale generation evidence with local Qwen3.5-9B. Each run targets five accepted examples in each planned category.

| Export artifact     | Examples | FinBench | SNB   | Train | Dev | Test |
|---------------------|----------|----------|-------|-------|-----|------|
| Live full benchmark | 3,000    | 2,000    | 1,000 | 2,408 | 296 | 296  |

Table 9: Accepted live full benchmark package with stable IDs, gate metadata, result samples, statistics, and manifest hash 8bc79a53a06b291a.

The full-run failure taxonomy shows where generation effort was spent before export: 1,335 rejected candidates came from diversity or duplicate controls during recovery, 374 from empty execution results, 66 from judge semantic rejection, and only 2 from schema invalidity. This supports the claim that deterministic schema gates made invalid Cypher rare, while refresh-scale generation mostly needs better value/template exploration.

For judge calibration, the released artifacts include an 80-row audit packet sampled from accepted and rejected full-run candidates, with 40 judge accepts, 40 judge rejects, 48 FinBench rows, 32 SNB rows, and all eight categories represented. The calibration tooling tracks graph, category, difficulty, strategy, and judge-decision coverage, and reports agreement, precision/recall, specificity, balanced accuracy, and Cohen’s kappa once human labels are filled. The current draft treats these labels as pending human review rather than as a generation gate.

Finally, we ran a downstream Text2Cypher evaluation using local Qwen3.5-9B on the exported full test split. The model saw schema text and the natural-language question, generated Cypher, and was evaluated by live execution against the corresponding FinBench or SNB database. The result verifies that the exported artifact can drive downstream model evaluation and gives a difficulty signal: the zero-shot 9B baseline solves simple aggregation examples but struggles on more operational categories.

## 7 Industry Use

PIPE-Cypher is intended to be rerun when an enterprise graph changes. The generated artifact records the schema snapshot, graph profile, model identifier, validation gates, execution samples, judge scores, difficulty features, and source run for every example. Long-running jobs can be monitored from JSONL records and recovered with category-specific top-up runs that reject questions accepted in earlier runs. This design supports private benchmark refreshes without exposing schemas or values to paid APIs, while still leaving audit hooks for data owners to sample accepted and rejected examples.

## 8 Conclusion

PIPE-Cypher provides a practical path for enterprises to create private, executable, balanced NL-to-Cypher benchmarks. The pipeline combines schema introspection, constrained generation, deterministic validation, execution feedback, diversity control, and automated judge review.

| Artifact property                   | Value                     |
|-------------------------------------|---------------------------|
| Categories                          | 8 balanced categories     |
| FinBench/category                   | 250                       |
| SNB/category                        | 125                       |
| Difficulty split                    | 1,521 easy / 1,479 medium |
| Unique labels / rel. types          | 7 / 9                     |
| Read/syntax/schema/exec/judge gates | 3,000/3,000               |

Table 10: Distribution and gate summary for the exported full benchmark artifact.

| Split          | Examples | Parse | Schema | Exec. success | Exec. acc. | Answer F1 |
|----------------|----------|-------|--------|---------------|------------|-----------|
| Live full test | 296      | 0.959 | 0.905  | 0.622         | 0.189      | 0.189     |

Table 11: Downstream Text2Cypher evaluation for local Qwen3.5-9B on the exported full benchmark test split.

## Limitations

Execution validity does not guarantee semantic correctness. LLM judges can over-accept plausible but wrong examples, so the final study includes a small human audit for calibration. Generated artifacts may contain private schema details or sampled values, so organizations should apply normal data governance controls.

## Ethics Statement

PIPE-Cypher is designed for private benchmark generation under local-model deployment constraints. Organizations using it should restrict benchmark artifacts to authorized users, review sampled values for sensitive content, and document whether judge calibration relied on human annotators.

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## A Additional Results



| Metric                                  | Value                 |
|-----------------------------------------|-----------------------|
| Question Distinct-1                     | 0.041                 |
| Question Distinct-2                     | 0.088                 |
| Question self-BLEU-2 (sampled)          | 0.826                 |
| Unique query-signature ratio            | 0.010                 |
| Top query-signature share               | 0.083                 |
| Category normalized entropy             | 1.000                 |
| Graph-category normalized entropy       | 0.980                 |
| Difficulty normalized entropy           | 1.000                 |
| Label coverage                          | 0.412                 |
| Relationship-type coverage              | 0.375                 |
| Property-name coverage                  | 0.407                 |
| Aggregation / negation / ordering rates | 0.500 / 0.125 / 0.125 |

Table 12: Diversity diagnostics for the full exported benchmark. Distinct-n follows text-generation usage; self-BLEU is lower when questions are less redundant.

| Failure bucket              | Count | Share of rejected |
|-----------------------------|-------|-------------------|
| Diversity/duplicate control | 1,335 | 0.751             |
| Empty result                | 374   | 0.210             |
| Judge semantic reject       | 66    | 0.037             |
| Schema invalid              | 2     | 0.001             |

Table 13: Failure taxonomy over full-run generation candidates before benchmark export. Accepted examples are excluded from the bucket shares.

| Audit packet property | Value   |
|-----------------------|---------|
| Rows                  | 80      |
| Judge accept / reject | 40 / 40 |
| FinBench / SNB rows   | 48 / 32 |
| Easy / medium rows    | 26 / 54 |
| Labeled rows          | 0       |

Table 14: Post-hoc judge calibration packet coverage. Human labels are pending and are not used as a generation gate.

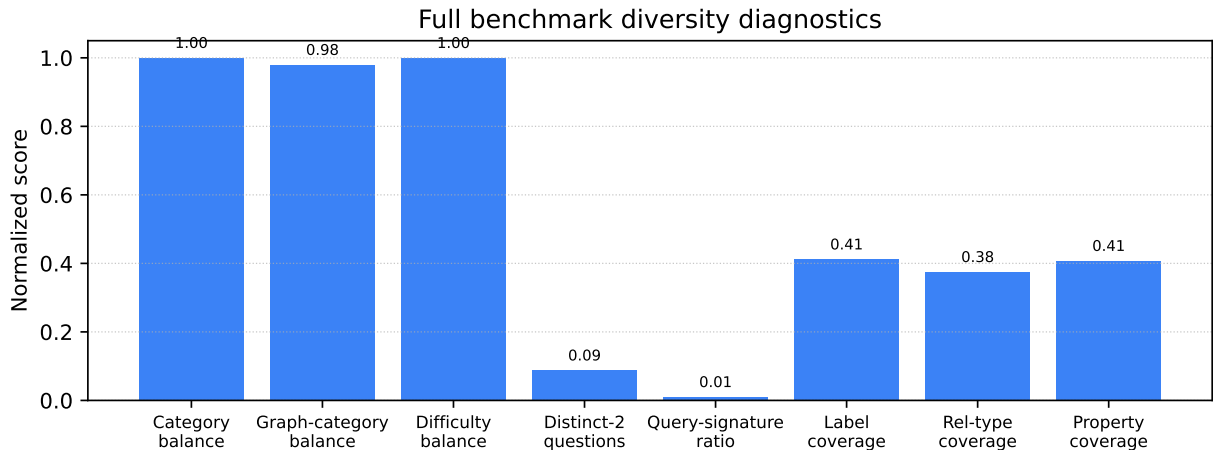


Figure 1: Diversity diagnostics for the full 3,000-example export. Category, graph-category, and difficulty balance are strong by construction; schema coverage and query-signature diversity expose remaining concentration from the fallback seeded-template run.

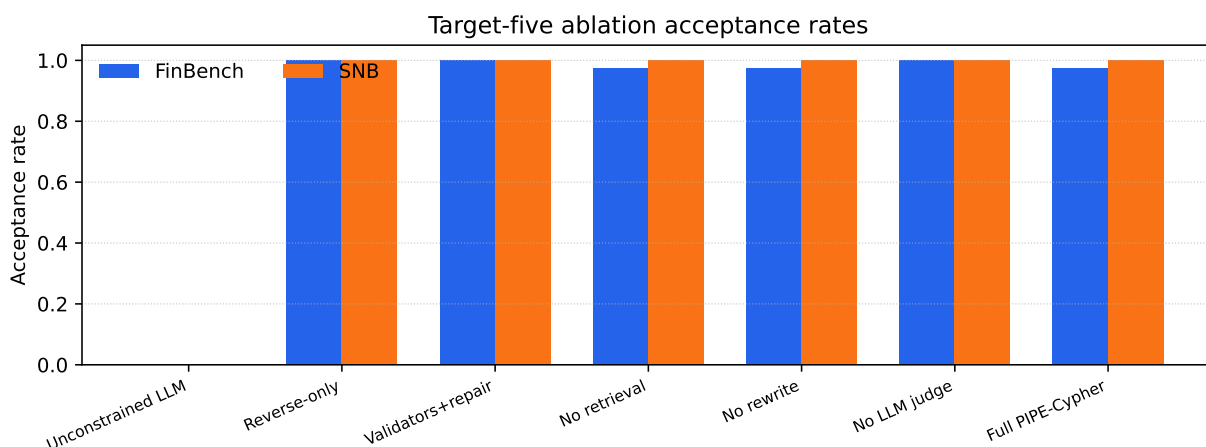


Figure 2: Target-five ablation acceptance rates on live FinBench and SNB graphs. The strict unconstrained local-LLM setting produces no usable examples, while reverse grounding and deterministic structure make the small target saturate across later variants.

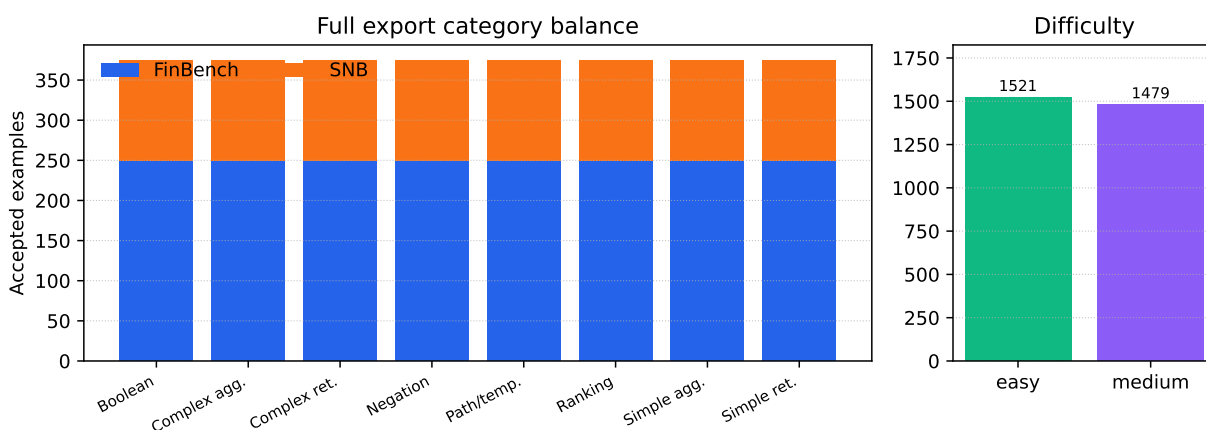


Figure 3: Full 3,000-example export distribution. The benchmark preserves the planned 2:1 FinBench/SNB graph mix while balancing all eight Cypher workload categories and maintaining a near-even easy/medium difficulty split.

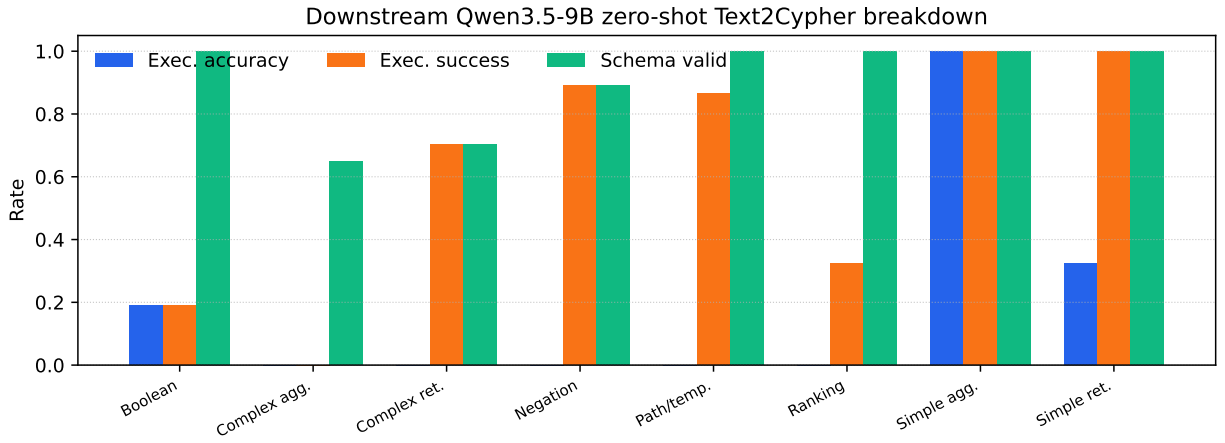


Figure 4: Downstream zero-shot Qwen3.5-9B Text2Cypher evaluation by workload category on the full test split. The breakdown separates final execution accuracy from parse/schema/execution success signals, making difficult operational categories visible.

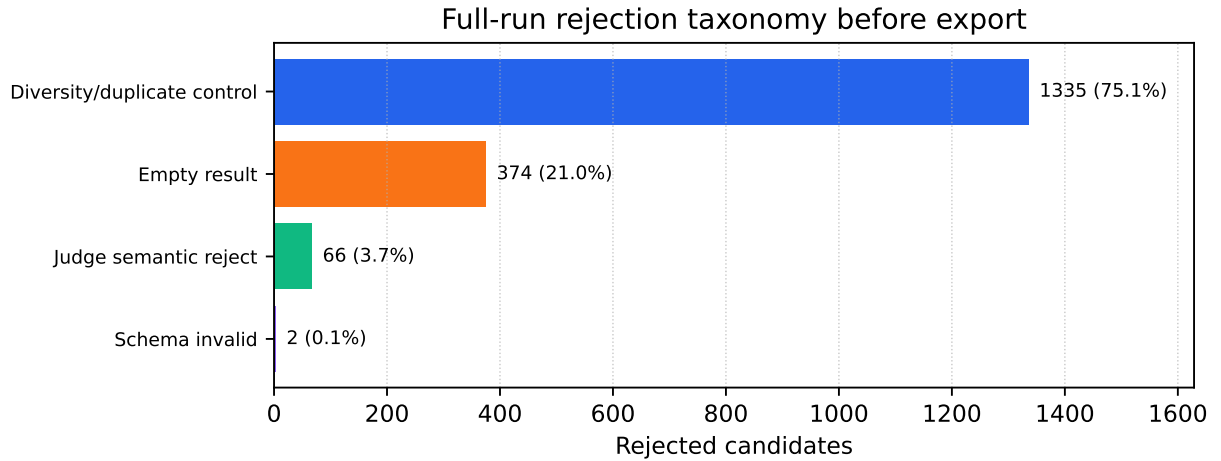


Figure 5: Full-run rejection taxonomy before benchmark export. Most rejected candidates come from duplicate/diversity control during category recovery and from empty execution results; schema-invalid Cypher is rare after deterministic governance.